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Enhanced Charge Transport in Hybrid Perovskite Field-Effect Transistors via Microstructure Control

*Andrew M. Zeidell, Colin Tyznik, Laura Jennings, Chuang Zhang, Hyunsu Lee, Martin Guthold, Z. Valy Vardeny, and Oana D. Jurchescu**

Hybrid organic–inorganic perovskites have recently gained immense attention due to their unique optical and electronic properties and low production cost, which make them promising candidates for a wide range of optoelectronic devices. But unlike most other technologies, the breakthroughs witnessed in hybrid perovskite optoelectronics have outgrown the basic understanding of the fundamental material properties. For example, the effectiveness of charge transport in relation to film microstructure and processing has remained elusive. In this study, field-effect transistors are fabricated and evaluated in order to probe the nature and dynamics of charge transport in thin films of methylammonium lead iodide. A dramatic improvement is shown in the electrical properties upon solvent vapor annealing. The resulting devices exhibit ambipolar transport, with room-temperature hole and electron mobilities exceeding 10 cm^2 V^{-1} s^{-1}. The remarkable enhancement in charge carrier mobility is attributed to the increase in the grain size and passivation of grain boundaries via the formation of solvent complexes.

1. Introduction

Hybrid organic–inorganic perovskites (HOIPs) have recently become one of the most promising contenders for the next-generation lightweight and flexible photovoltaic (PV) solar cell platforms. HOIPs such as methylammonium lead trihalide ($CH_3NH_3PbX_3$, referred to hereafter as $MAPbX_3$, where X may be I, Cl, Br, or a combination) are based on earth-abundant materials and display intriguing intrinsic properties such as tunable bandgap,[1] high absorption coefficient,[2] large photoluminescence quantum yield,[3] high charge carrier mobilities,[4–6] and long carrier diffusion lengths.[7] These properties, coupled with their compatibility with solution processing, have inspired significant research on the perovskite materials. Solar cells based on HOIPs have witnessed an unprecedented growth in efficiency in only a few years, faster than any PV technology to date, with power conversion efficiencies greater than 20%.[8] In addition, HOIPs have been incorporated in photodetectors,[9] optically pumped lasers with tunable wavelength,[10] and high-efficiency light-emitting diodes.[3] Remarkably, all these devices were manufactured at temperatures below or around 100 °C.

The small electron effective mass calculated for HOIPs suggests that the charge carrier mobilities, μ, may be very high in absence of defects and other scattering sites.[11,12] Indeed, mobilities exceeding 100 cm^2 V^{-1} s^{-1} were determined in single crystals from space charge limited current (SCLC) and terahertz (THz) spectroscopy measurements.[6,7] The mobilities obtained from field-effect transistors (FETs), however, have been much lower; a maximum value of around 1 cm^2 V^{-1} s^{-1} was reported in $MAPbX_3$ perovskite FETs,[13–15] and a maximum of around 2 cm^2 V^{-1} s^{-1} in $Cs_x(MA_{0.17}FA_{0.83})_{1-x}Pb(Br_{0.17}I_{0.83})_3$ triple cation perovskites,[16] despite significant effort dedicated to this topic.[17–22] The discrepancy between the measured mobilities with various techniques may arise from the fact that in FETs the interface properties are evaluated, which are prone to defects, whereas the other methods access the bulk properties. On the other hand, recent work suggests that the high mobility values in hybrid perovskites may be an overestimation since the charge transport in these compounds is inhibited due to the existence of a plethora of processes such as polaron formation,[23] scattering from acoustic and optical phonons,[5,11,24] ionic motion, polarization disorder of the organic cations, and dynamic disorder.[14,18,25] A room-temperature mobility of ≈200 cm^2 V^{-1} s^{-1} was predicted in $MAPbI_3$ in the presence of phonon scattering alone, with no other factors that limit charge carrier transport.[11] It is critical to assess the true intrinsic electrical properties of hybrid perovskites to advance their use in commercial optoelectronic applications. FETs are excellent experimental platforms to access such information with minimal or no approximations necessary. In addition to being the fundamental unit in a wide range of electronic applications, FET devices allow for a basic study of the factors limiting charge transport in perovskites, with a great control over the charge density in the semiconductor layer. Unfortunately, because of the challenges associated with

A. M. Zeidell, C. Tyznik, L. Jennings, H. Lee, Dr. M. Guthold, Dr. O. D. Jurchescu
Department of Physics and Center for Functional Materials
Wake Forest University
Winston-Salem, NC 27109, USA
E-mail: jurchescu@wfu.edu
Dr. C. Zhang, Dr. Z. V. Vardeny
Department of Physics and Astronomy
University of Utah
Salt Lake City, UT 84112, USA
The ORCID identification number(s) for the author(s) of this article can be found under https://doi.org/10.1002/aelm.201800316.

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device fabrication, access to such measurements is rather limited. Room-temperature device operation in HOIPs is known to be hampered by the ion migration, which screens electronic transport, and by the structural fluctuations and dynamical disorder caused by rotation of the organic cations around their axis and PbX_6 octahedral distortions;[18,25] therefore, only a limited number of reports on working FET devices exist.[13,14,19,22] For a comprehensive review on the recent advancements on metal-halide perovskite transistors, we direct the reader to the manuscript by Anthopoulos and co-workers.[26]

In the present work, we controlled the structure and morphology of the $MAPbI_{3-x}Cl_x$ HOIP layer through deposition and postprocessing, and obtained ambipolar FETs with electron and hole mobilities exceeding 10 $cm^2 V^{-1} s^{-1}$ at room temperature. We note that greater mobilities were reported in tin-based perovskites, $(C_6H_5C_2H_4NH_3)_2SnI_4$.[27] To understand the origin of the enhanced charge transport as a result of processing, we studied the structural changes as a function of solvent annealing time, and related the results to the electrical properties of the film. Solvent annealing techniques have been previously used to increase grain size, improve crystallization, and enhance performance in perovskite thin films.[28–30] Here, we demonstrate over ten times improvement in electron and hole average mobilities, respectively, upon exposure to *N*,*N*-dimethylformamide (DMF) solvent at room temperature.

2. Results and Discussion

The crystal structure of the hybrid lead-halide perovskites is shown in **Figure 1**a. The methylammonium cation, the lead cation, and the halide anion are shown separately in the right panel. The perovskite films were deposited by spin-coating from a precursor solution that consisted of PbI_2, $PbCl_2$, and CH_3NH_3I (MAI) in DMF solvent as seen in Figure 1b; the details are provided in the "Experimental Section." To perform the thermal conversion from precursor to perovskite film, the substrates were heated on a hot plate in the glovebox, without exposure to air. During thermal annealing (Figure 1c), Cl^- ions assist in the nucleation of $MAPbI_3$ crystals by reacting with MAI, while liberating the excess $CH_3NH_3^+$ via the rapid formation of a short-lived $MAPbCl_3$ phase. This process slows down the crystallization of the film through the sublimation of volatile MACl, that results in larger crystal domains and a final Cl^- concentration of <3%.[31] The $MAPbI_{3-x}Cl_x$ perovskite films were exposed to DMF solvent vapors at room temperature in increments of 20 min (Figure 1d). Several other solvents were tested, but we found that they did not have significant effects on the film morphology and electrical performance (see the "Experimental Section").

The as-cast perovskite films exhibited a discontinuous morphology that consists of small crystalline grains (**Figure 2**a), which is typical for films produced from the mixed halide precursor.[31] The exposure of the films to solvent vapors (at room temperature and controlled time intervals), i.e., solvent vapor annealing, resulted in a morphology with long needle-like grains spanning tens of micrometers (Figure 2b), but the film reorganization also led to more severe dewetting from the substrate. Note that the entire width of the source and drain electrodes was considered in the mobility estimation; thus, the values we report are underestimations. Figure 2c shows an atomic force microscopy (AFM) image of a pristine perovskite film before solvent annealing; the film displays a rough surface, with small grain size and an irregular grain structure. In contrast, the AFM image of a solvent

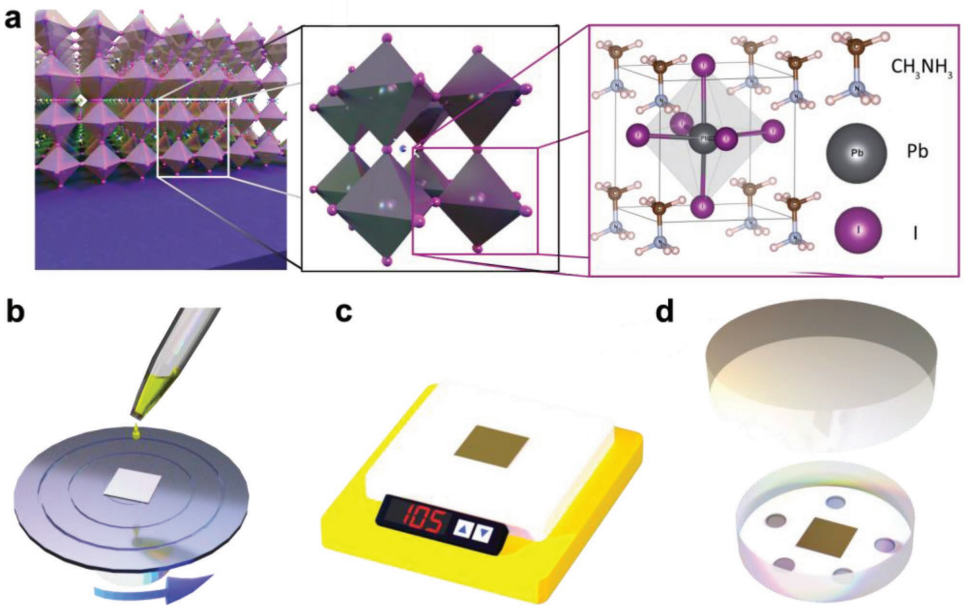

Figure 1. Perovskite film structure, deposition, and solvent annealing treatment. a) Chemical structure of $MAPbI_3$. The methylammonium cation is shown in the upper right-hand corner, while the lead cations are shown by silver spheres, and iodides are shown by purple spheres. b) Perovskite are deposited via spin coating from a precursor solution, and c) thermally converted at 105 °C. d) Solvent annealing is performed by exposing the thermally converted film to a solvent atmosphere in an enclosed Petri dish by placing five 10 µL droplets around the film, with exposure taking place in increments of 20 min. All steps were performed in an inert N_2 atmosphere.

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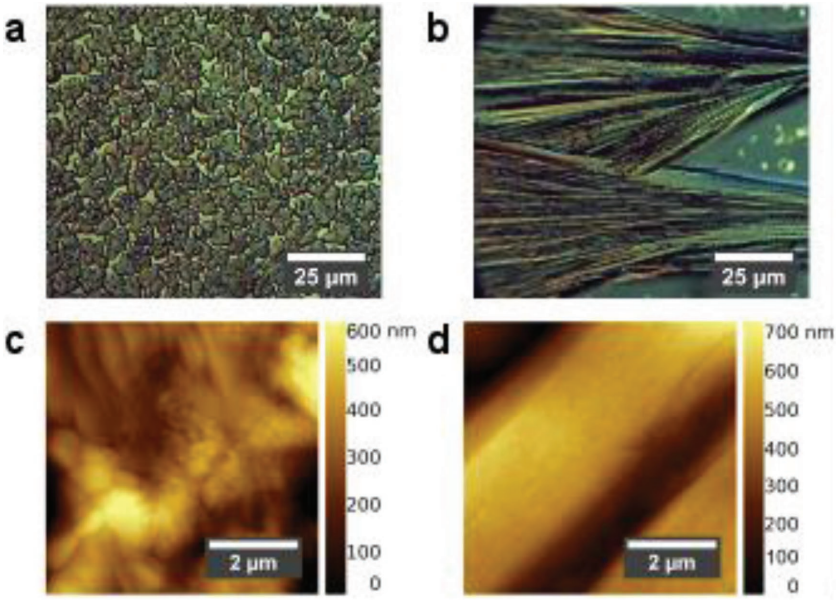

Figure 2. Optical and AFM images of pristine and solvent annealed perovskite films. a) Optical microscopic image of the pristine perovskite film in the channel region. b) The film in the FET channel after 20 min of solvent exposure; note the formation of large needle-shaped crystals. c) AFM image of pristine film, displaying nonuniform surface characteristics. d) AFM image of solvent-annealed film displaying highly uniform grain microstructure.

annealed film indicates that these films exhibit significantly larger grains and smoother surfaces (Figure 2d). These conclusions are also supported by the scanning electron microscopy (SEM) images presented in Figure S1 (Supporting Information).

Field-effect transistor devices were fabricated on films that represent each stage in the solvent annealing process. We adopted a top-contact, bottom-gate architecture, with Au source and drain contacts and SiO_2 dielectric (**Figure 3**a). Representative source–drain current versus gate–source voltage (I_D–V_{GS}) characteristics for both hole and electron transport in the saturation regime (drain–source voltage, $V_{DS} = -40$ V) are shown in Figure 3b and Figure 3c, respectively. An example of an output curve is depicted in Figure S2 (Supporting Information), and its deviation from linearity at low source–drain voltages, as well as nonideal saturation regime reached, suggests that the contact resistance is quite severe in these devices. The FET characterization was carried out by performing the measurements in the "sweep-mode," and the mobility estimation was performed during the forward sweep of the transfer curves, in order to minimize the contributions due to ionic motion. Similar measurements taken in "pulsed-mode" may reduce the contribution to transport of the ionic motion which is ubiquitous in these materials, as reported by Senanayak et al.[14] For devices that showed a linear dependence of the sqrt (I_D) on the applied V_{GS}, such as the examples included in **Figure 4**a,b, the standard models for FET characterization were used for mobility determination (details are provided in the "Experimental Section"). The devices with Schottky contacts exhibited a "kink" in the transfer characteristics, as shown in Figure 4c,d. This nonideal behavior results from non-Ohmic injection, which produces a very large contact resistance. Since the contact resistance is reduced with increasing V_{GS}, access to the intrinsic mobility is only possible at higher voltages.[32,33] In order to avoid the mobility overestimation in our devices, in this case we performed the linear fit in the high-voltage regime (see the red line).

The as-cast films (not exposed to solvent) exhibited ambipolar transport, with hole mobility $\mu_h = 1.8 \pm 0.3$ cm^2 V^{-1} s^{-1} and electron mobility $\mu_e = 0.7 \pm 0.2$ cm^2 V^{-1} s^{-1}. Not all devices exhibited ambipolar behavior, most likely due to scattering at the interface with the gate dielectric and contact effects. We have not been able to obtain working devices in perovskite FETs at room temperature with SiO_2 dielectric and bottom source and drain contacts, in agreement with previous reports;[17,18] although we are aware that this is possible upon illumination.[19] This may originate from the higher contact resistance characteristic to this geometry, and possible degradation of the electrode surface through interaction with the perovskite precursor solution. By simply examining the band diagram of this system, electron transport is not expected to occur with gold contacts.[34] The FET measurements, nevertheless, have undoubtedly proven that both hole and electron transport can be distinguished in devices with gold contacts, a result found by several groups.[16,17,19] This behavior suggests that the energy band diagram is not sufficient for predicting the injection

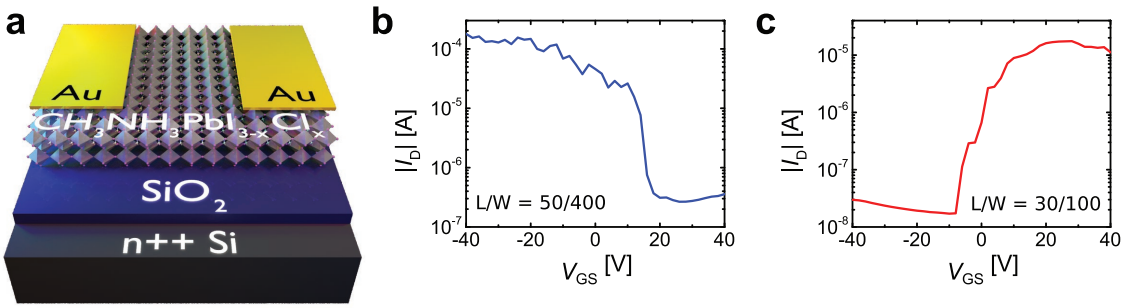

Figure 3. FET geometry and representative electrical characteristics of perovskite devices. a) Top-contact, bottom-gate perovskite FET device geometry. Representative plot of I_D–V_{GS} characteristics of perovskite FET ($V_{DS} = \pm 40$ V) for b) holes and c) electrons, respectively. The values of the channel lengths and widths are included in the inset.

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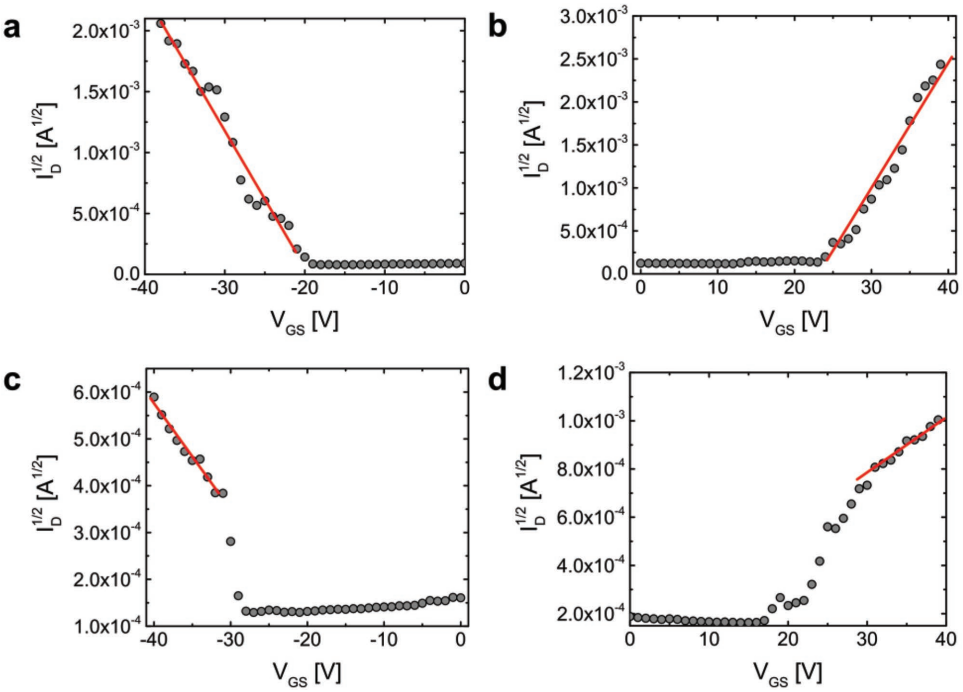

Figure 4. Example transfer curves showing linear fits for mobility estimation. a) Hole and b) electron sqrt (I_D) versus V_{GS} curve showing a linear dependence, consistent with the standard models used for mobility determination. c) Hole and d) electron nonideal curves, where the two slopes are the result of gated contacts. The linear fits used in the mobility calculations are shown in a red line.

process in the HOIP semiconductors, based simply on the alignment of the metal work function with the semiconductor's energy level involved in charge transport, because the processes that occur at the semiconductor/metal interface may be dominant.

The change in the hole and electron mobilities as a function of solvent annealing time is shown in **Figure 5**a and Figure 5b, respectively. A gradual increase in mobility is observed for both electrons and holes, reaching average mobilities of $\mu_h = 10 \pm 2.5$ cm^2 V^{-1} s^{-1} and $\mu_e = 10 \pm 3.4$ cm^2 V^{-1} s^{-1} at 80 min solvent annealing time ("maximum" mobilities were measured to be μ_h = 15.8 cm^2 V^{-1} s^{-1} and μ_e = 15.7 cm^2 V^{-1} s^{-1}, and a histogram of the hole and electron mobilities at this optimum annealing condition is included in Figure S3 in the Supporting Information). Further solvent annealing to 100 min or longer yielded a decrease in charge carrier mobility of both holes and electrons, and also led to nonoperational devices. We also evaluated the impact of solvent annealing on other device parameters and the results are presented in Figure S4 (Supporting Information). It can be observed that the subthreshold slope decreases with increasing the annealing time, suggesting that the lowering of the number of trap densities at the perovskite/SiO_2 interface is responsible for the improvements in the mobility. The on/off ratio, on the other hand, remains constant under the conditions explored in this study.

To elucidate the factors that mediate the improved device performance upon solvent annealing, we performed X-ray diffraction (XRD) measurements on pristine and solvent annealed films

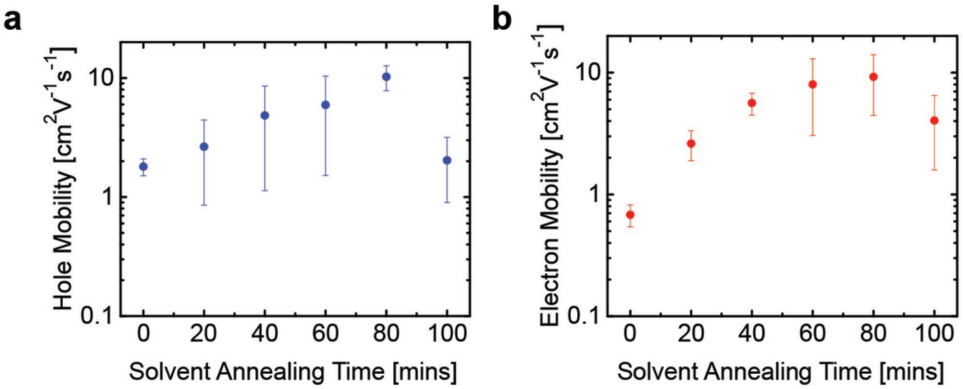

Figure 5. Average charge carrier mobilities as a function of solvent annealing time. a) Hole and b) electron mobilities for FET devices measured after exposure to DMF solvent vapor at various times, t. Note that the average hole and electron mobilities reach maximum at $t = 80$ min, before decreasing. Error bars indicate the standard deviation from the mean mobility value.

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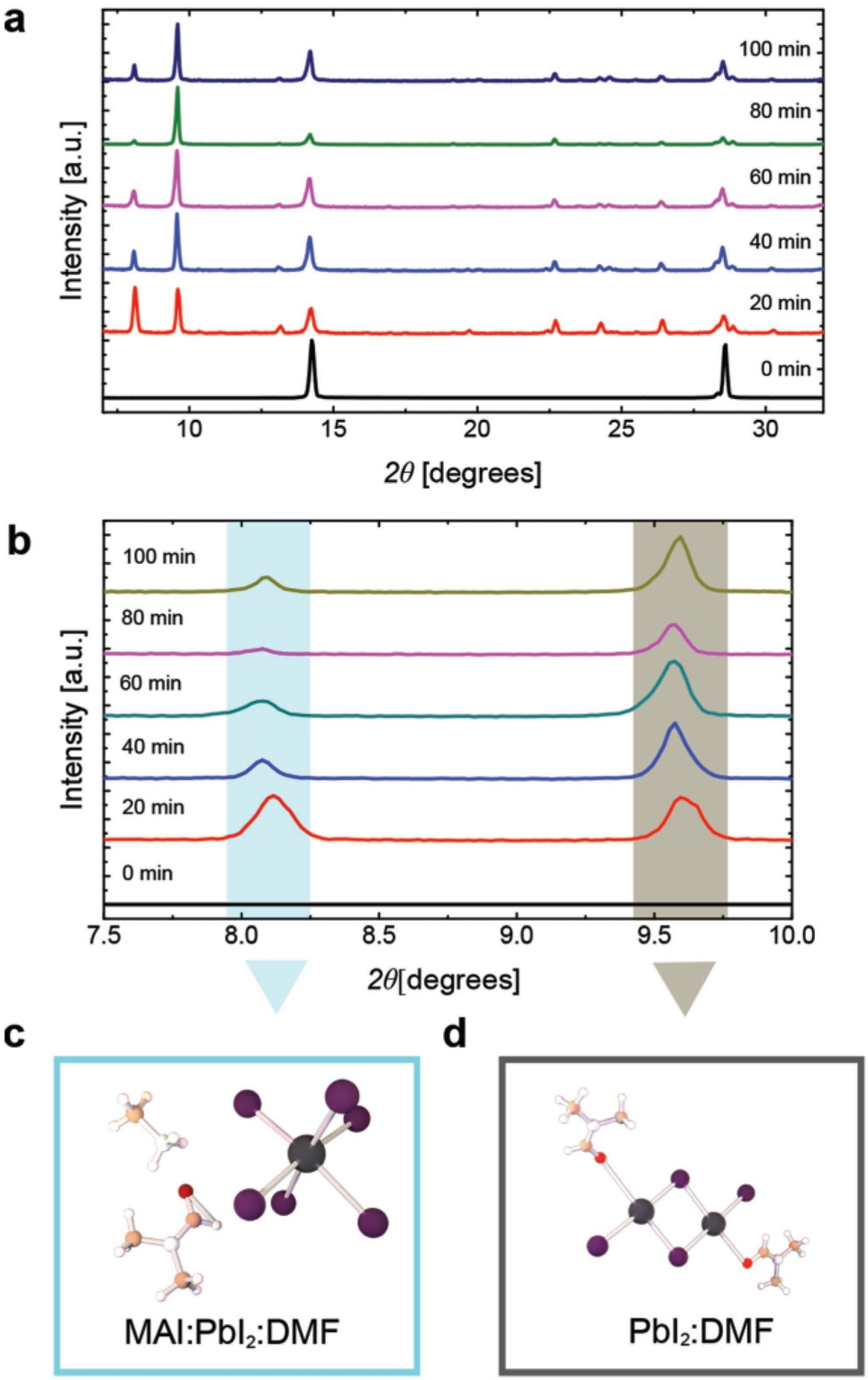

Figure 6. XRD characterization of films and identification of solvates. a) XRD spectra of perovskite films at different stages of solvent annealing. b) XRD peaks that correspond to c) MAI:PbI_2:DMF solvate and d) PbI_2:DMF ligand.

for each exposure time. The results are included in **Figure 6**a. It can clearly be seen that the as-cast film crystallizes in the known tetragonal phase that adopts a preferential [110] orientation, where the XRD peaks located at 14.12° and 28.44° correspond to the (110) and (220) reflections, respectively.[35] After 20 min of exposure to solvent vapors, new peaks are present in the XRD

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spectra indicating the loss of preferential crystalline orientation. Two additional peaks appear at 8.09° and 9.57° (Figure 6b), confirming the formation of the MAI:PbI_2:DMF solvate[35] and the PbI_2:DMF ligand,[28] respectively (Figure 6c,d). We were able to assign all XRD peaks in the spectrum as coming from the tetragonal phase and the aforementioned solvate and ligand bands; thus, no other $MAPbI_3$ polymorphs were formed.

Solvent annealing of the perovskite films had several important consequences: removal of preferential orientation, increase in grain size, and formation of intermediate species. Whereas the [110] orientation in the pristine film is preferred for photovoltaic applications,[15,36] our results indicate that this direction is not favorable for charge transport in an FET architecture. This is not surprising since the transport in the PV cells occurs in a sandwich structure, perpendicular to the film surface, whereas the contacts in FETs are coplanar. We note that the removal of the preferential orientation occurs within the first 20 min of solvent annealing; additional exposure to solvent vapor did not alter the number of XRD peaks, suggesting that no major transformation happens in the solid-state order. The exposure to solvent likely enabled the perovskite to partially redissolve, with the molecules rearranging during the subsequent crystallization process, while also allowing for chemical reactions to occur. The reaction between the residual MAI, PbI_2, and DMF solvent leads to the formation of the MAI:PbI_2:DMF solvate, whereas that between the PbI_2 and solvent yielded the PbI_2:DMF ligand.[37] PbI_2-rich grain boundaries in the pristine film can introduce midgap states,[38] whereas the solvent complexes increase the performance in photovoltaic devices.[39] This is important because in 3D perovskites, there is no pathway for unreacted components in the film to fully form into perovskites, leaving PbI_2 and MAI residues in the film as contaminants.[40] They accumulate at the grain boundaries, leading to potential barriers that inhibit charge transport, though they do not change the nonradiative recombination dynamics in PV devices.[41] Their solvation into the MAI:PbI_2:DMF solvate and/or PbI_2:DMF ligand consumes unreacted materials from the pristine film, thus "healing" the grain boundaries, and consequently enhancing charge transport. This increases the charge carrier mobility, in agreement with our results shown in Figure 4. The formation of the PbI_2:DMF ligand causes an expansion of the (001) plane of PbI_2, detected as a shift of the PbI_2 12.66° peak to 9.57°. This creates a favorable condition for the formation of new perovskite crystals via gradual intercalation of MAI at room temperature, decomposing the MAI:PbI_2:DMF solvate and explaining the gradual disappearance of the corresponding XRD peak. Following 80 min of solvent annealing, there is a competing effect given by the microstructure change and dewetting of the film, which results in device performance degradation (see Figure S5 in the Supporting Information).

3. Conclusion

In conclusion, we have succeeded in tuning the grain size and grain-boundary chemistry in lead-iodide perovskite field-effect transistors by solvent annealing. We showed that this process can lead to a significant morphological change in which the original preferential orientation of the thin film was removed; the grain size was increased drastically and the grain boundaries were healed through doping with intermediate phases based on solvent complexes with unreacted material. As a result, we demonstrated high-performance ambipolar perovskite FETs that exhibit mobilities greater than 10 $cm^2\ V^{-1}\ s^{-1}$ at room temperature. We combined structural and electrical studies to clarify the mechanism through which solvent annealing improves charge transport in perovskite FETs. This work has allowed us to gain insight into the impact of film morphology on charge transport in perovskite materials, an important milestone toward achieving high-performance optoelectronic devices such as FETs, photovoltaics, light-emitting diodes, and photodetectors.

4. Experimental Section

Perovskite Film Deposition: $CH_3NH_3PbI_{3-x}Cl_x$ films were deposited from a precursor solution by spin coating. To fabricate this solution, PbI_2 (99.999% trace metals basis, Aldrich), $PbCl_2$ (99.999% trace metals basis, Aldrich), and CH_3NH_3I (98%, Aldrich) were mixed in *N*,*N*-dimethylformamide solvent (anhydrous, 99.8%, Sigma-Aldrich) at a 3:0.9:0.3 MAI:$PbCl_2$:PbI_2 molar ratio, with a final concentration of 38 wt%. The solution was stirred overnight at 80 °C. To fabricate the devices, the process was started from heavily n++ doped Si wafers with a thermally grown SiO_2 layer of 300 nm, which were cleaned sequentially in hot acetone, hot isopropanol, and UV–ozone in 10 min steps, followed by a 30 s deionized water (DI) rinse and blown dry with nitrogen. The substrate was then baked at 155 °C prior to transferring it to an inert atmosphere glovebox (<0.1 ppm H_2O, <0.1 ppm O_2). Next, the precursor solution was deposited via spin coating, using a 10 s ramp to a spinning speed of 3000 rpm. The perovskite layer was formed by annealing the obtained film for 90 min at 105 °C inside the glovebox.

Solvent Annealing of Perovskite Films: The perovskite films obtained upon thermal annealing of the precursor films were quenched to room temperature and placed in a Petri dish, in the same glovebox. About 50 µL of *N*,*N*-dimethylformamide (anhydrous, 99.8%, Sigma-Aldrich) was added to the same container, deposited in five even droplets around the sample using an Eppendorf pipette with disposable polypropylene tip, and the container was covered with the glass lid, but not sealed. The films were solvent annealed at room temperature in 20 min steps, and removed from the Petri dish immediately to avoid contact with nonevaporated solvent. Next, they were immediately placed under slight vacuum at 100 Torr for 30 min. Different films were evaluated for each annealing time investigated in this study. Organic solvents such as acetone, toluene, and the polar aprotic solvent dimethyl sulfoxide were also used as solvents, but it was found that they did not produce operational devices.

XRD of Perovskite Films: 2θ scans of perovskite films deposited onto SiO_2 were obtained using an X-Ray diffractometer (Bruker Desktop D2 Phaser). The step size for the 2θ scans was 0.02° in an interval of 7°–50°.

AFM of Perovskite Films: Scans were obtained using an Asylum MFP-3D-Bio AFM (Asylum Research, USA) at ambient conditions. A rectangular silicon cantilever in tapping mode (Olympus, AC160TS-R3, Spring constant k = 8.4–57 N m^{-1}, resonance frequency f = 200–400 kHz) was used.

FET Fabrication and Electrical Characterization: Top-contact, bottom-gate field-effect transistors were fabricated using the films described above, in which the heavily n++-doped Si played the role of gate electrode and the SiO_2 layer was the gate dielectric. 60 nm Au source and drain electrodes were deposited via thermal evaporation through a shadow mask in a Kurt Lesker Spectros system with a base pressure of at least 10^{-7} Torr. Devices were characterized using an Agilent 4155C Semiconductor Parameter Analyzer in dark and a vacuum of at least 10^{-4} Torr. The mobility was calculated from the slope of the square

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root of the drain current with respect to the gate–source voltage curve, using Equation (1)

$$I_D = \frac{W}{L}\frac{C_i}{2}\mu(V_{GS} - V_T)^2 \quad (1)$$

where L and W are the channel length and the channel width, respectively, and C_i is the capacitance per unit area of the dielectric. For FETs that showed deviations from this model, the mobility evaluation was always performed at higher voltages to minimize the contribution from contact effects.

SEM Measurements: Perovskite films were characterized with cold field emission SEM using a JEOL JSM-6330F instrument with an accelerating voltage of 10.0 kV.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

charge transport, field-effect transistors, metal-halide perovskites, mobility, solvent annealing

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